

spin statistics (ch 12)

- spin stat. then follows from symmetrization
invariance (Lec.)

12.1 Identical particles

- creation operators
→ multi particle state

$$| \dots s_1 \vec{p}_1 u_1 \dots s_2 \vec{p}_2 u_2 \dots \rangle =$$

$$\dots \sqrt{2u_1} a_{\vec{p}_1 s_1 u_1}^\dagger \dots \sqrt{2u_2} a_{\vec{p}_2 s_2 u_2}^\dagger \dots |0\rangle$$

$$\langle s_1 \vec{p}_1 u_1 \dots | s'_1 \vec{p}'_1 u'_1 \dots \rangle =$$

$$\prod_i \delta_{u_i - u'_i} \delta_{s_i, s'_i} 2u_i (2\pi)^3 \delta^3(\vec{p}_i - \vec{p}'_i)$$

- line counting
in diff
order

→

$$|s_{n_1, p_1, u_1} \dots s_{n_l, p_l, u_l}\rangle = \dots \sqrt{a_{n_l}} a_{p_l, u_l}^\dagger \dots \sqrt{a_{n_1}} a_{p_1, u_1}^\dagger \dots |0\rangle$$

- same phys. state

→ Differ only by phase

$$(\dots s_{n_1, p_1, u_1} \dots s_{n_l, p_l, u_l})$$

$$= \alpha | \dots s_{n_1, p_1, u_1} \dots s_{n_l, p_l, u_l}$$

$$u/\alpha = e^{i\varphi}$$

- α can't depend on pos

? - no 1-dim rep. of loc. group

2nd P Boeck

$$| \dots s_1 \vec{p}_1 u \dots s_2 \vec{p}_2 u \rangle = \hat{\alpha}_u | \dots s_1 \vec{p}_1 u \dots s_2 \vec{p}_2 u \rangle$$

$$\alpha_u = \pm 1$$

- $\alpha_u = 1$ bosons
boson-eigenstate relativistic

$\alpha_u = -1$ fermions (Fermi Dirac statistics)

Boson:

$$a_{\vec{p}_1 s_1 u}^{\dagger} a_{\vec{p}_2 s_2 u}^{\dagger} |4\rangle = a_{\vec{p}_2 s_2 u}^{\dagger} a_{\vec{p}_1 s_1 u}^{\dagger} |4\rangle$$

$$\therefore [a_{\vec{p}_1 s_1 u}^{\dagger}, a_{\vec{p}_2 s_2 u}^{\dagger}] = 0$$

$$\langle \vec{p}_1 | \vec{p}_2 \rangle = 2u_1 (2\pi)^3 \delta(\vec{p}_1 - \vec{p}_2)$$

$$\rightarrow \left\{ a_{\vec{p}_1, s_1, u}, a_{\vec{p}_2, s_2, u}^\dagger \right\} = (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2) \delta_{s_1, s_2}$$

Fermion:

$$\left\{ a_{\vec{p}_1, s_1, u}, a_{\vec{p}_2, s_2, u}^\dagger \right\} = 0$$

$$\left\{ a_{\vec{p}_1, s_1, u}, a_{\vec{p}_2, s_2, u} \right\} = (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2) \delta_{s_1, s_2}$$

- Fermions in same state:

$$a_{\vec{p}}^\dagger a_{\vec{p}}^\dagger |0\rangle = -a_{\vec{p}}^\dagger a_{\vec{p}}^\dagger |0\rangle$$

$$= 0$$

- Pauli exclusion

12.2 spin & total fermi path Dep.

- Interchange particles physically

$\phi = 0$ particles at

u_1 & u_2

- later at u_1 & u_2

$$S(\phi_1(u_1)\phi_2(u_2)) \equiv e^{i\phi_2} |\phi_2(u_1)\phi_1(u_2)\rangle$$

- swap

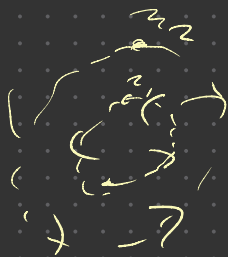


$\phi = 0$

no exchange



$\phi = \pi$ exchange



$$\phi = 2\pi \text{ no ending } c$$

$$- n \in \mathbb{Z} \text{ since } \gamma(\phi) = \exp(i\phi)$$

$$\therefore \text{ if } \phi = \pi \text{ no ending } c$$

$$S | \phi_1(u_1) \phi_2(u_2) \rangle$$

$$= \pm | \phi_2(u_1) \phi_1(u_2) \rangle$$

- only fermions & bosons

- $\rightarrow$ more by Poisson $\rightarrow$ generating
- (- nothing useful under time (definition))
- Rot. dependence $\rightarrow$ $\rightarrow$

scalar =

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \langle \phi_2(x_2) \phi_1(x_1) \rangle$$

Non-trivial Dirac:

$$\Delta_S(\theta_2) = \begin{pmatrix} e^{i\pi/2\theta_2} & & & \\ & e^{i\pi/2\theta_2} & & \\ & & e^{i\pi/2\theta_2} & \\ & & & e^{i\pi/2\theta_2} \end{pmatrix}$$

$\theta_2 \rightarrow \pi$ has i & $-i$

suppose γ spin $\in$ (out of page)

$$\psi_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = i\psi_1$$

$$\Rightarrow S | \psi_1(r_1) \psi_2(r_2) \rangle$$

$$= - | \psi_2(r_1) \psi_1(r_2) \rangle$$

- definition spin

$$r \perp p_{\text{spin}}$$

$$S | \psi_1(r_1) \psi_2(r_2) \rangle = \epsilon i \phi_{\text{spin}} | \psi_2(r_1) \psi_1(r_2) \rangle$$

can have $n \neq 1$

(anyons)

$$S(r_1)$$

9. Dirac Quantization

recall:

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + b_p^\dagger e^{ipx})$$

$$\phi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p^\dagger e^{-ipx} + b_p e^{ipx})$$

Lag. (Mink)

$$\mathcal{L} = \bar{\Psi}(i\not{\partial} - m)\Psi$$

$$(i\not{\partial} - e\not{A} - m)\psi = 0$$

$$\bar{\Psi}(i\not{\partial} - e\not{A} - m) = 0$$

written in terms of

$$u(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_S \\ \sqrt{p \cdot \bar{\sigma}} \xi_S \end{pmatrix}$$

$$v(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta_S \\ -\sqrt{p \cdot \bar{\sigma}} \eta_S \end{pmatrix}$$

$$u / \xi_1 = \eta_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\cancel{u} / \xi_2 = \eta_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

const.

$$\rightarrow \mathcal{H} = \sum_S \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{E_p}} \left(a_{p\mu}^\dagger \tilde{u}_\mu^S e^{-ipx} + b_{p\mu}^\dagger \tilde{v}_\mu^S e^{ipx} \right)$$

$$\bar{\mathcal{H}} = \sum_S \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{E_p}} \left(u_\mu^\dagger \bar{a}_\mu^S e^{ipx} + \bar{b}_\mu^\dagger \bar{v}_\mu^S e^{-ipx} \right)$$

12.4 9-matt. comm.

$$\{ \{ a_p(t) a_q(t') \} \} = -T \{ a_q(t') a_p(t) \}$$

Fermionic time ordering:

$$T \{ \psi(x) \chi(y) \}$$

$$= \psi(x) \chi(y) \theta(x_0 - y_0)$$

$$- \chi(y) \psi(x) \theta(y_0 - x_0)$$

again:

$$\langle 0 | \psi(x) \psi(y) | 0 \rangle$$

$$= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{i\omega_p} \frac{1}{i\omega_q}$$

$$\times \langle 0 | (a_p^\dagger e^{ipx} + b_p e^{-ipx}) (a_q + b_q^\dagger) | 0 \rangle$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{e^{-ipx}}{2\omega_p} + i \vec{p} \cdot \vec{x}$$

$$\langle 0 | \phi(0) \phi^*(n) | 0 \rangle$$

$$= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-i\omega_p t - i\vec{p} \cdot \vec{n}}$$

Time order
 $\rightarrow$ Feynman propagator

$$\langle 0 | T \{ \phi^*(n) \phi(0) \} | 0 \rangle$$

$$= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\omega_p} \left[e^{i\vec{p} \cdot \vec{n} - i\omega_p t} \theta(t) + e^{-i\vec{p} \cdot \vec{n} + i\omega_p t} \theta(-t) \right]$$

$$\rightarrow \langle 0 | T \{ \phi^*(n) \phi(0) \} | 0 \rangle$$

$$= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip \cdot n}$$

(L.T.)

If anticom., not C.T.

$$\left(\int \frac{d^3 p}{(2\pi)^3} \frac{u}{\sqrt{p^2 + m^2}} \frac{-i}{p^2 + m^2} e^{i p x} \right)$$

12.4.2 9/14/09!

$$\langle 0 | \psi(0) \bar{\psi}(x) | 0 \rangle = \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \frac{1}{\sqrt{2m_p}} \frac{1}{\sqrt{2m_q}}$$

$$+ \sum_{s, s'} (a_p^s \bar{u}_p^s + b_p^{s'} \bar{v}_p^{s'}) (a_q^{s'} \bar{u}_q^{s'} e^{-i q x} + b_q^s \bar{v}_q^s e^{-i q x}) | 0 \rangle$$

$$= \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \frac{1}{\sqrt{2m_p}} \frac{1}{\sqrt{2m_q}} \sum_{s, s'} \bar{u}_p^s \bar{u}_q^{s'} (a_p^s \bar{u}_p^s + b_p^{s'} \bar{v}_p^{s'}) (a_q^{s'} \bar{u}_q^{s'} + b_q^s \bar{v}_q^s) e^{-i q x}$$

$$= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2m_p} \sum_s \bar{u}_p^s \bar{u}_p^s e^{i p x}$$

$$(\psi(0) \bar{\psi}(0) \text{ is a constant})$$

$$\sum_{s \in 1/2} u_s(p) \bar{u}_s(p) = \not{p} + m$$

$$\sum_{s \in 1/2} v_s(p) \bar{v}_s(p) = \not{p} - m$$

$$\begin{aligned} & \Rightarrow \langle 0 | \bar{\psi}(x) \psi(0) | 0 \rangle \\ &= \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \\ & \quad \times \left(\langle 0 | b_q \bar{v}_q - \bar{v}_q' e^{-iqx} + a_q' \bar{u}_q' e^{iqx} \right) \\ & \quad \times (a_p^\dagger u_p + b_p^\dagger v_p) / q \end{aligned}$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-ipx} (\not{p} - m)$$

$$= (i\not{\partial} - m) \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-ipx}$$

Alas! from before

$$\langle 0 | \psi(0) \bar{\psi}(x) | 0 \rangle$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ipx} (\not{p} + m)$$

$$= (i\not{\partial} + m) \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ipx}$$

$$\rightarrow \langle 0 | \psi + \epsilon \psi(0) \bar{\psi}(x) \psi(0) | 0 \rangle$$

$$= (i\not{\partial} - m) \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \left[e^{i\vec{p}\vec{x} - i\omega_p t} \theta(t) - e^{-i\vec{p}\vec{x} + i\omega_p t} \theta(-t) \right]$$

$$= (-i\cancel{\not{p}} + m) \int \frac{d^4 p}{(2\pi)^4} \frac{i}{\sqrt{p^2 + m^2} p^2 - m^2 + i\epsilon} e^{ipx}$$

Not L.I.

→ need Anti comm.

$$\rightarrow \langle 0 | T \bar{\psi}(x) \psi(x) \rangle = (-i\cancel{\not{p}} + m) \int \frac{d^4 p}{(2\pi)^4} \frac{i}{\omega^2 - \vec{p}^2 + i\epsilon} e^{ipx}$$

Spinor Feynman Propagators

$$\langle 0 | T \bar{\psi}(x) \psi(x) \rangle = \int \frac{d^4 p}{(2\pi)^4} \frac{i(\cancel{\not{p}} + m)}{p^2 - m^2 + i\epsilon} e^{ipx}$$

